

METROPOLITAN COMPREHENSIVE
CANCER CENTER

Department of Pathology & Precision Medicine
1400 University Health Plaza, Suite 400 · Molecular Diagnostic Laboratory

CLIA License #39D2049182
CAP Accredited #7192840
(fictional identifiers — sample only)

Integrated Pathology & Molecular Diagnostic Report SPECIMEN ID: SAMPLE-0000000

A PATIENT DEMOGRAPHICS & SPECIMEN METADATA	
PATIENT NAME	PATIENT ID / MRN
Eleanor Vance (fictional)	PT-8820491
DATE OF BIRTH	ORDERING PHYSICIAN
11/14/1972 (Age 53)	Dr. Robert Sterling, MD — Oncology
GENDER	REPORTING PATHOLOGIST
Female	Dr. Sarah Jenkins, MD, FCAP
SPECIMEN TYPE	COLLECTION DATE
FFPE Tissue (Formalin-Fixed Paraffin-Embedded)	July 18, 2026
PRIMARY SPECIMEN SOURCE	REPORT DATE
Liver Metastasis — Core Biopsy	July 24, 2026

B CLINICAL DIAGNOSIS & HISTOPATHOLOGY SUMMARY

Invasive Breast Carcinoma — Infiltrating Ductal Carcinoma, Triple-Negative Phenotype

Histologic Grade: Grade 3 (Poorly Differentiated)

Clinical Staging: Stage IV — cT2, N1, M1 (distant liver & lymph node involvement)

ECOG Performance Status: 1 — Restricted in strenuous activity, fully ambulatory

RECEPTOR	RESULT	METHOD
ER (Estrogen Receptor)	Negative <1%	IHC
PR (Progesterone Receptor)	Negative <1%	IHC
HER2/neu	Negative — IHC 1+, FISH Non-Amplified	IHC / FISH

C NEXT-GENERATION SEQUENCING — GENOMIC PROFILE



GENE	VARIANT / ALTERATION	ALLELE FREQ.	CLASSIFICATION	CLINICAL SIGNIFICANCE
BRCA1	c.5266dupC (p.Gln1756Profs*74)	48.2% (Germline)	Pathogenic	PARP inhibitor / platinum sensitivity
PIK3CA	c.3140A>G (p.His1047Arg)	22.4% (Somatic)	Pathogenic	PI3Kα inhibitor target
TP53	c.818G>A (p.Arg273His)	51.0% (Somatic)	Pathogenic	Tumor suppressor loss

TUMOR MUTATIONAL BURDEN

11.4 Mut/Mb

TMB-High

MICROSATELLITE STATUS

Stable (MSS)

Stable

PD-L1

Combined Positive Score (CPS) = 2

D TREATMENT HISTORY



AUG 2025 — COMPLETED

Neoadjuvant Chemotherapy

Dose-dense AC-T: Doxorubicin, Cyclophosphamide → Paclitaxel



SEP 2025

Surgical Resection

Left total mastectomy + axillary lymph node dissection



NOV 2025 – JUN 2026 — DISCONTINUED

1st-Line Metastatic Therapy

Olaparib 300 mg BID (PARP inhibitor); discontinued due to progression in liver metastases

E CURRENT LABORATORY & ORGAN FUNCTION PANELS as of Jul 28, 2026

PARAMETER	RESULT	REFERENCE RANGE	STATUS
Absolute Neutrophil Count (ANC)	2.4 ×10 ³ /μL	1.5 – 8.0	Adequate
Platelet Count	185 ×10 ³ /μL	150 – 450	Normal
Hemoglobin	11.2 g/dL	12.0 – 15.5	Grade 1 Anemia
eGFR (Kidney Function)	74 mL/min/1.73m ²	> 60	Normal Renal Function
Serum Creatinine	0.92 mg/dL	0.59 – 1.04	Normal
ALT / AST (Liver Function)	42 / 38 U/L	< 35	Mild Elevation
CNS Status (Contrast Brain MRI)	No active brain metastases	—	Negative for CNS Lesions

IMPRESSION

Tumor exhibits a pathogenic germline BRCA1 frameshift mutation alongside an acquired PIK3CA H1047R hotspot mutation following progression on first-line PARP inhibitor therapy (Olaparib).

RECOMMENDATION

Patient demonstrates adequate renal, hematologic, and hepatic baseline function without active CNS disease. Highly suitable for targeted Phase I–III clinical trial enrollment evaluating novel PI3K/AKT/mTOR inhibitors, antibody–drug conjugates (ADCs), or combination immunotherapy agents.

Dr. Sarah Jenkins, MD, FCAP

Director of Molecular Diagnostics

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